Flow of control

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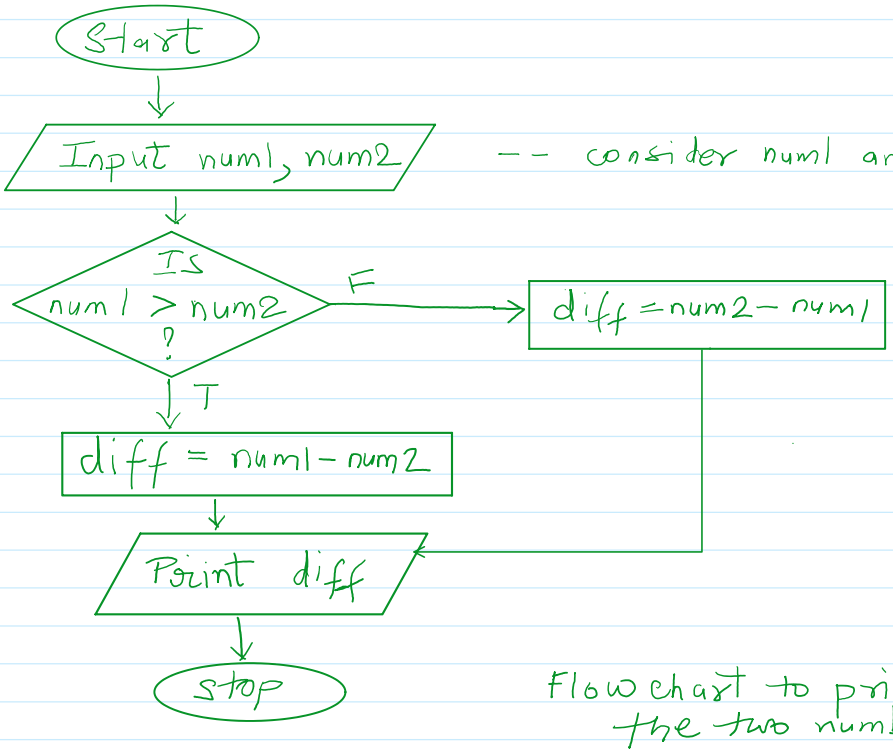
Selection

Now suppose we have rupees 10 to buy a pen. on visiting the stationary shop there are varieties of pen priced at Rupees 10 each. Here we have to decide which pen to buy. Similarly, when we use the directional services of a digital map to reach home from one place to another, we notice that it shows more than one path like the least crowded path or short test distance path, Etc. We decide the path as per our priority. A decision involves selecting from one of the two or more possible options. in programming, this concept of decision making or selection is implemented. is implemented with the help of **if… else…** statement.



Flowchart to print positive difference of the two numbers num1 and num2.

The syntax of if statement :

```
if condition :
    statement(s)
```

The syntax of if... else statement:

```
if condition :
    statement(s).
else:
    statement(s).
```

The syntax of if... elif... else statement

```
if condition :
    statement(s)
elif condition :
    statement(s)
elif condition :
```

```
elif condition :
    statement(s)
elif condition :
    statement(s)
...
else :
    statement(s)
```

WAP to print the positive difference of two numbers.

```
num1 = int(input("Enter first number: "))
num2 = int(input("Enter second number: "))

if num1 < 0 or num2 < 0 :
    print("Error! numbers cannot be negative")
else:
    if num1 > num2 :
        diff = num1 - num2
    else:
        diff = num2 - num1
    print("The difference of", num1, "and", num2, "is", diff)
```

Write a program to create a simple calculator to perform basic arithmetic operations on two numbers. The program should do the following:

- Accept two numbers from the user.
- Ask user to input any of the operator. (+, -, *, /, //(lower division), %(remainder), ^(exponentiation)). An error message is displayed if the user enters anything else.

Write a program to input three angles of a triangle and check whether its construction is possible or not. If possible, then check and display whether it is an acute angle triangle, right angled or an obtuse angle triangle. Otherwise, display a "triangle is not possible".

```
# Write a program to input three angles of a
# triangle and check whether its construction is
# possible or not. If possible, then check and
# display whether it is an acute angle triangle,
# right angled or an obtuse angle triangle.
# Otherwise, display a "triangle is not possible".

a = float(input('Enter first angle: '))
b = float(input('Enter second angle: '))
c = float(input('Enter third angle: '))
sum = a + b + c
if a == 0 or b == 0 or c == 0 or sum != 180:
    print('Triangle is not possible')
else:
    print('Triangle is possible')
    if a == 90 or b == 90 or c == 90 :
```

```
        print('Right Angled Triangle')
    elif a > 90 or b > 90 or c > 90 :
        print('Obtuse Angled Triangle')
    else:
        print('Acute Angled Triangle')
```